

End-to-End Robotic Vision and Control System Using Doosan Manipulator, Intel RealSense, and RoboFlow

Presented by Group 4

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Abstract: A Comprehensive Robotic Manipulation Pipeline

This project details a complete pipeline for robotic manipulation, integrating advanced robot control, precise camera calibration, and robust object detection into a unified, deployable system. We cover three core components:

Robot Control

Focuses on controlling a Doosan robotic arm within an Ubuntu environment, utilizing a custom GUI and dedicated software to leverage its capabilities.

Vision Calibration

Employs an Intel RealSense camera and chessboard-based calibration to establish accurate spatial relationships, addressing practical challenges like focus and stability.

Object Detection

Utilizes RoboFlow for dataset preparation, model training, and deployment, enabling real-time object detection for enhanced robotic tasks.

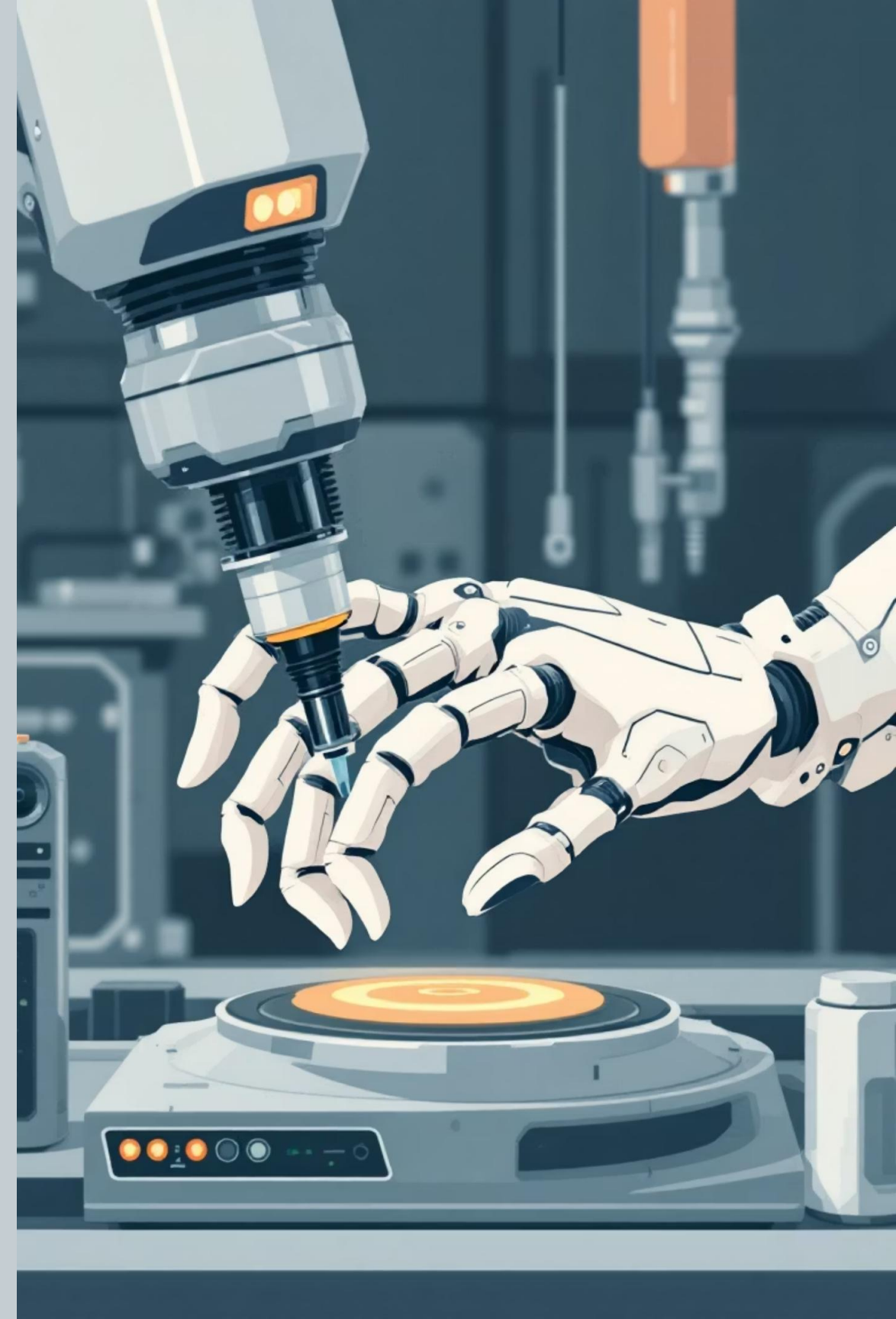
Together, these elements demonstrate a modular approach to vision-guided robotics, highlighting both implementation details and real-world system limitations.

Problem Statement: Bridging the Gap in Robotic Manipulation

Despite significant advancements, robotic manipulators continue to face challenges in complex, multi-stage tasks that demand:

- **Precise Visual Identification:** Accurately locating operation-specific regions within varied environments.
- **Line-Guided Motion:** Generating smooth, precise motion trajectories based on visual cues.
- **Multi-Arm Coordination:** Synchronizing actions between multiple robotic systems for collaborative tasks (though not a primary focus of this project, it represents a common challenge).
- **Real-time Integration:** Seamlessly combining perception, trajectory generation, and manipulation in dynamic scenarios.

This project specifically addresses these critical gaps by developing a unified pipeline designed to detect task-specific operational lines using RoboFlow, convert these visual detections into executable robot paths, and perform sequential manipulation tasks.



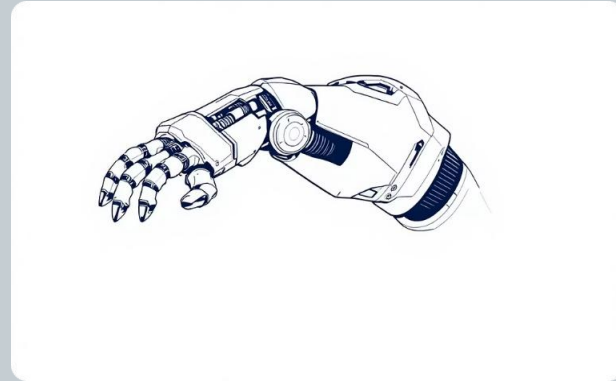
Our Comprehensive Technology Stack



Operating Environment

Ubuntu Linux: The robust foundation for our system.

ROS 2 middleware (rclpy, colcon): Facilitating inter-process communication and development.



Robot Control

Doosan A0509 Collaborative Robot: Our primary manipulator.

DSR_ROBOT2 APIs (movej, movei, posj): For intricate motion control.

RViz: For 3D visualization and simulation.



Programming & GUI

Python 3: The core language for system development.

Tkinter-based custom Teach Pendant GUI: For intuitive human-robot interaction.

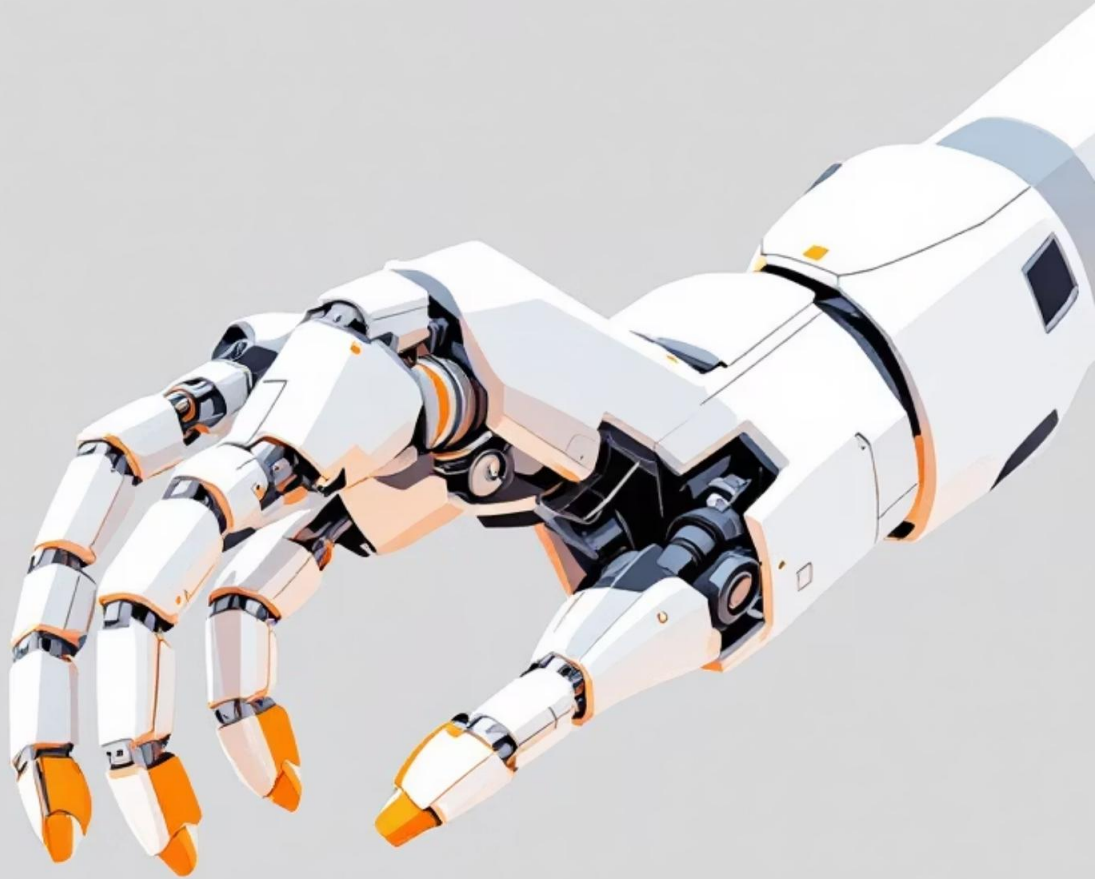


Vision & AI

Intel RealSense RGB-D Camera: For depth and color perception.

Chessboard-based camera calibration: Ensuring accuracy.

RoboFlow: For advanced object detection and model management.



Robot Motion & Control: Precision and Safety



Motion Control Fundamentals

Joint-space control (`movej()`): For direct joint angle manipulation, ideal for initial positioning and obstacle avoidance.

Cartesian linear motion (`move1()`): For straight-line movements in 3D space, essential for precise interaction with objects.



Teach Pendant GUI

A custom Tkinter-based GUI provides:

- Intuitive **joint and Cartesian jogging**.
- **Waypoint teaching and replay** for complex trajectories.
- **Speed and loop control** for customizable task execution.



Safety & Autonomous Execution

Operating in **autonomous robot mode** ensures unattended operation.

Motion locking during execution prevents unintended movements and enhances safety during critical operations.

Camera Calibration: Achieving Spatial Accuracy

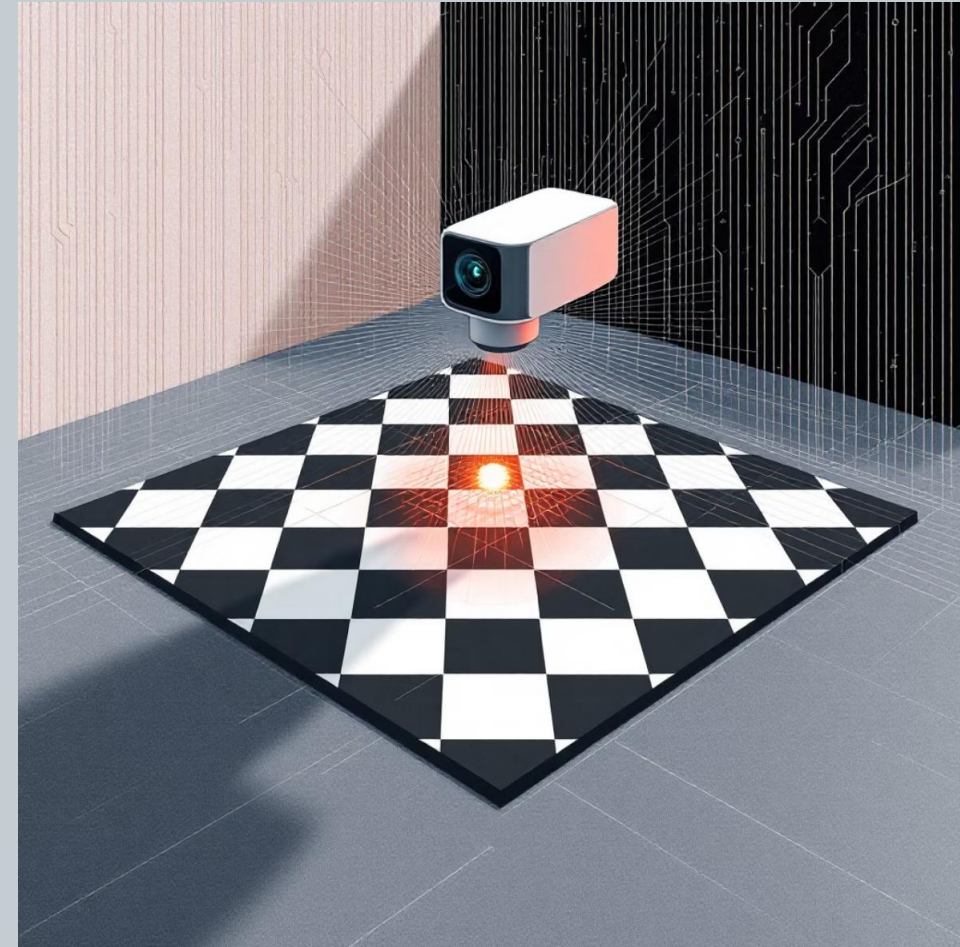
Accurate spatial understanding is critical for robotic manipulation. Our calibration process ensures the robot can precisely interpret visual data.

Camera Setup

- **Intel RealSense RGB-D camera:** Selected for its integrated depth and color capabilities.
- **Fixed view of robot workspace:** Ensuring a consistent perspective for optimal calibration and operation.

Calibration Technique

- **Chessboard-based intrinsic calibration:** To correct lens distortions and determine internal camera parameters.
- **Camera-to-robot extrinsic estimation:** To define the precise spatial relationship between the camera and the robot's base frame.



📄 Challenges Encountered

- **Focus loss:** A significant issue when the target moves outside the camera's optimal field of view.
- **Lighting and distance sensitivity:** External factors heavily influence calibration accuracy and stability, necessitating controlled environments.



Object Detection with RoboFlow: Empowering Robotic Vision

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Dataset Preparation

- **Image collection and annotation:** Meticulously gathered and labeled for task-specific objects.
- **Dataset versioning and augmentation:** Ensuring data quality and increasing model robustness through varied examples.

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Model Training & Deployment

- **Model training and validation:** Iterative process to achieve high accuracy and generalization.
- **Exported for real-time inference:** Optimized models deployed for swift, on-the-fly detection.

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System Role

- **Provides object location and class:** Crucial data for robotic decision-making.
- **Guides robot task execution:** Directing the manipulator based on detected object characteristics.

End-to-End System Integration: From Vision to Action



Integrated Pipeline

- **Camera capture** for real-time visual data.
- **Object detection** identifying targets.
- **Coordinate transformation** translating pixels to robot movements.
- **Robot motion planning and execution** to complete tasks.



Architecture Benefits

- **Modular and scalable design:** Allowing for easy expansion and adaptation to new tasks.
- **Real-time operation:** Essential for dynamic industrial and research applications.



Key Limitations

- **Calibration sensitivity:** External factors can impact precision.
- **Open-loop execution:** Current setup lacks continuous feedback for error correction, posing challenges in highly variable environments.

This project successfully demonstrates a comprehensive, vision-guided robotic system, highlighting both its capabilities and areas for future refinement.

